

firm regardless of AI exposure. For workers aged 22-25, we find a 15 log-point decline in relative employment for the most AI-exposed quintiles compared to the least exposed quintile, a large and statistically significant effect. Estimates for other age groups are much smaller in magnitude and not statistically significant. These findings imply that the employment trends we observe are not driven by differential shocks to firms that employ a disproportionate share of AI-exposed young workers.

Fifth, the labor market adjustments are visible in employment more than in compensation. In contrast to our findings for employment, we find little difference in annual salary trends by age or exposure quintile, suggesting possible wage stickiness. If so, AI may have larger effects on employment than on wages, at least initially, or even that AI may boost wages for as many workers as it hurts.

Sixth, the above facts are largely consistent across sample constructions designed to address various alternative explanations for the core findings. Our results are not driven solely by computer occupations or by occupations susceptible to remote work and outsourcing. We also find that the AI exposure taxonomy did not meaningfully predict employment outcomes for young workers further back in time, before the widespread use of LLMs, including during the unemployment spike driven by the COVID-19 pandemic. The patterns we observe in the data appear most acutely starting in late 2022 and early 2023, around the time of rapid proliferation of generative AI tools.³ They hold for both occupations with a high share of college graduates and ones with a low college share, suggesting deteriorating education outcomes during COVID-19 do not drive our results. For non-college workers, we find evidence that experience may serve as less of a buffer to labor market disruption, as low college share occupations exhibit divergent employment outcomes by AI exposure up to age 40.

While we explore a variety of alternative explanations, we caution that the facts we document may in part be influenced by factors other than generative AI. Taken as a whole, our results are consistent with the hypothesis that generative AI has begun to affect entry-level employment. We intend to continue to track the data on an ongoing basis to assess whether these trends change in the future.

³In particular, OpenAI introduced ChatGPT in November 2022. It was reported to have over 100 million active users by January 2023 and 1.7 billion website visitors in October 2023 (DeVon, 2023).